CNN Report: Forest Fire Detection

1. Introduction: Forest fires pose a significant threat to ecosystems, wildlife, and human settlements. Early detection of forest fires is crucial for timely intervention and mitigation efforts. In this report, we present the development and evaluation of a Convolutional Neural Network (CNN) model designed to detect forest fires in satellite images. The CNN leverages deep learning techniques to automatically classify images as either containing a forest fire or not.

2. Problem Statement: The primary objective of this project is to create a CNN model capable of accurately identifying forest fires in satellite/drone images. This task involves binary classification, where the model must categorize images as either 'forest fire' or 'no forest fire.' The model's performance will be evaluated based on metrics such as accuracy, precision, recall, and the Area Under the ROC Curve (AUC).

3. Dataset: The dataset consists of satellite images capturing both wildfire instances and non-fire scenarios. It is divided into three subsets: training, validation, and testing. The training set contains 30,000 images, while the validation and testing sets contain 6,000 images each. Each image is labeled according to the presence or absence of a forest fire.

4. Research Methods:

4.1 Data Collection:

- The dataset was collected from satellite imagery sources and curated to include images of forest fires and non-fire scenes. The dataset is published on Kaggle.

4.2 Data Preprocessing:

- Images were resized to a standard size of 200x200 pixels to ensure uniformity and reduce computational complexity.
- Data augmentation techniques such as rotation, shifting, shearing, zooming, brightness adjustment, and horizontal

flipping were applied to increase the diversity of the training dataset.
- Generators were created for training, validation, and testing, with batch size set to 32 and images rescaled to the range [0, 1].
- These variables were set according to an extensive grid search performed to determine their best possible values.

4.3 Model Architecture Design:

- A CNN architecture was designed to extract features from input images and perform binary classification.
- The architecture consists of convolutional layers followed by max-pooling layers for feature extraction, followed by fully connected dense layers for classification.
- Rectified Linear Unit (ReLU) activation functions introduce non-linearity, aiding in learning complex patterns.
- The output layer consists of a single neuron with a sigmoid activation function for performing the binary classification task.
- The model's architecture was developed using pre-trained CNN's architectures, which were later tested using fire image datasets and modified to the specific task. Using this techniques, the model achieved satisfactory results in the binary classification task.

4.4 Model Compilation:

- The CNN model was compiled using the binary cross-entropy loss function, suitable for binary classification tasks.
- The Adam optimizer was employed for efficient gradient descent optimization.
- Evaluation metrics including accuracy, AUC, and recall were specified to monitor the model's performance during training.

4.5 Model Training:

- The model was trained on the augmented training dataset and validated on the validation set.

- Early stopping was implemented to prevent overfitting, with training halting if the validation loss did not improve for 6 consecutive epochs.

4.6 Model Evaluation:

- The trained model was evaluated using the test dataset to assess its performance on unseen data.
- Evaluation metrics including loss, accuracy, AUC, precision, recall, and F1-score were computed.
- A classification report was generated to provide a comprehensive overview of the model's performance.

5. Results:

- The CNN model achieved high performance in accurately detecting forest fires in satellite images.
- Test results indicated a low test loss of 0.0609, high test AUC of 0.9789, and impressive test accuracy of 0.9977.
- Precision, recall, and F1-score metrics demonstrated the model's ability to effectively classify both positive and negative instances.

6. Conclusion: The developed CNN model can be used for early detection of forest fires in satellite imagery. By using deep learning techniques and careful model design, the model demonstrates robust performance metrics, making it a reliable solution for forest fire detection and mitigation efforts. Further improvements may focus on optimizing model hyperparameters and exploring additional data sources to enhance model generalization.

Additionally, visualization techniques will be employed to analyze model performance and gain insights into its decision-making process.

Model Summary:

```
super().__init__(
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 198, 198, 16)	448
max_pooling2d (MaxPooling2D)	(None, 99, 99, 16)	0
conv2d_1 (Conv2D)	(None, 97, 97, 32)	4,640
max_pooling2d_1 (MaxPooling2D)	(None, 48, 48, 32)	0
conv2d_2 (Conv2D)	(None, 46, 46, 64)	18,496
max_pooling2d_2 (MaxPooling2D)	(None, 23, 23, 64)	0
flatten (Flatten)	(None, 33856)	0
dense (Dense)	(None, 128)	4,333,696
dense_1 (Dense)	(None, 1)	129

```
Total params: 4,357,409 (16.62 MB)
Trainable params: 4,357,409 (16.62 MB)
Non-trainable params: 0 (0.00 B)
```